

Assignment 1 - Hard

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1 Instructions

This assignment is based on Lectures 1-6: bond features and risks, pricing and yield measures, duration and convexity, benchmark spreads and term structure, Treasury and corporate debt markets, and residential mortgages/MBS.

The questions are original CFA-style practice questions written from the course material. They are not copied from CFA Institute books or exams.

Questions 1-10 are conceptual, Questions 11-20 are calculations, and Questions 21-25 use the generated spreadsheet files in this folder.

1.1 Question 1 (Conceptual)

A CIO argues that the safest way to reduce risk is to buy the highest-yielding bonds with the shortest maturities. Give a deep critique.

Answer. Short maturity reduces exposure to level changes in rates, but high yield is often compensation for credit distress, weak liquidity, poor covenants, or near-term refinancing risk. A very short distressed bond can be more dangerous than a longer high-quality bond because the main risk is not duration; it is whether the issuer survives the maturity wall. The correct risk question is which factor is being paid for and whether the spread is enough after expected default loss, downgrade risk, liquidity cost, and portfolio constraints.

1.2 Question 2 (Conceptual)

Why can a bond's yield-to-maturity be a misleading measure of expected return?

Answer. YTM assumes promised cash flows are paid, coupons are reinvested at the same yield, and the bond is held to maturity. Those assumptions fail for callable bonds, mortgage-backed securities, distressed bonds, and investors with shorter horizons. Expected return must incorporate reinvestment rates, future sale yield, default probability, recovery, taxes, and embedded option exercise.

1.3 Question 3 (Conceptual)

Explain why convexity is economically valuable even if two bonds have the same duration and yield.

Answer. If two option-free bonds have the same duration and yield, the more convex bond has a more favorable response to rate volatility: larger gains when yields fall and smaller losses when yields rise. Convexity is valuable because it improves the distribution of price changes.

Investors often accept lower yield for more convexity, while callable mortgages may offer more yield because they sell convexity to borrowers.

1.4 Question 4 (Conceptual)

A portfolio has the same effective duration as its benchmark. Why can it still lose heavily relative to the benchmark when the curve twists?

Answer. Effective duration summarizes sensitivity to a parallel shift. A curve twist changes short, intermediate, and long rates by different amounts. If the portfolio is overweight long key-rate duration and underweight intermediate key-rate duration relative to the benchmark, the same total duration can hide a major mismatch. Key rate duration is needed to expose that curve-shape risk.

1.5 Question 5 (Conceptual)

Give an intuition for why a theoretical spot curve is a set of discount rates, not a set of expected future short rates.

Answer. Spot rates are today's prices for zero-coupon cash flows at each maturity. They embed time value, risk premiums, liquidity, financing specialness, and supply-demand effects. Forward rates extracted from spot rates are break-even rates under no-arbitrage pricing, but they are not pure forecasts unless risk premiums are zero and investors are risk-neutral.

1.6 Question 6 (Conceptual)

Why can a wider spread be rational for a more liquid corporate bond in stressed markets?

Answer. Normally liquidity lowers required yield. In stress, the most liquid bond may be the one investors sell first because it can actually trade, causing its spread to widen faster than illiquid bonds whose prices are stale. The observed spread can therefore reflect immediacy, balance-sheet pressure, and mark-to-market activity, not only fundamental credit deterioration.

1.7 Question 7 (Conceptual)

Explain the investor's hidden short option position in a mortgage pass-through.

Answer. The mortgage borrower owns the right to prepay. The MBS investor has effectively sold that option because principal returns when it is least convenient: faster when rates fall and reinvestment yields are low, slower when rates rise and the investor would prefer principal back. This produces negative convexity and makes duration state-dependent.

1.8 Question 8 (Conceptual)

Why is ‘risk risk’ especially relevant for structured fixed-income products?

Answer. Structured products can transform simple loans into tranches with nonlinear exposure to default timing, prepayments, volatility, liquidity, and correlation. A yield pickup may look attractive if the investor models only average loss or average life, while ignoring tail behavior and option exposure. Risk risk is the danger of owning a payoff the investor cannot actually explain under stress scenarios.

1.9 Question 9 (Conceptual)

How can bond covenants transfer value between shareholders and bondholders?

Answer. Shareholders benefit from risk-taking because upside is theirs while part of downside is borne by creditors. Negative covenants restrict actions such as additional debt, asset sales, or dividend leakage that would weaken bondholder claims. Strong covenants reduce agency conflict and can lower required spreads; weak covenants leave bondholders exposed to value transfer.

1.10 Question 10 (Conceptual)

Why is leverage in a bond portfolio conceptually different from simply buying longer-duration bonds?

Answer. Both can increase rate exposure, but leverage also introduces financing risk, margin calls, haircut changes, repo rollover risk, and forced-sale risk. A long-duration unlevered Treasury position can suffer mark-to-market losses, but a levered position can become insolvent or be liquidated at bad prices even if the long-run thesis is correct. The financing channel is therefore a separate risk factor.

1.11 Question 11 (Calculation)

A pension analyst studies an 8-year semiannual-pay corporate bond with \$1,000 face value and 4.75% coupon. The current yield is 5.25%. Reprice the bond when yield is 4.75% and 5.75%, then estimate empirical modified duration and convexity using a 50 bp shock.

Yield	Price
4.75%	\$1,000.00
5.25%	\$967.68
5.75%	\$936.59

Answer. Using the three prices, empirical modified duration = $(P_{\text{minus}} - P_{\text{plus}})/(2 \times P_0 \times 0.005) = 6.553$. Empirical convexity = $(P_{\text{minus}} + P_{\text{plus}} - 2P_0)/(P_0 \times 0.005^2) = 51.03$. The duration is the first-order exposure, while the convexity shows the curvature becomes material for larger shocks.

1.12 Question 12 (Calculation)

A liability-driven investor must pay \$2,700,000 in 5 years. Available zero-coupon STRIPS are:

Maturity	Spot rate
1 year	3.80%
3 years	4.25%
5 years	4.70%

Compute the present value of the 5-year liability using the correct rate.

Answer. The liability is a single 5-year cash flow, so it should be discounted at the 5-year spot rate, not an average curve yield. $PV = \$2,700,000/(1.047)^5 = \$2,146,003.15$.

1.13 Question 13 (Calculation)

A 4-year annual-pay corporate bond has a 3.50% coupon and \$1,000 face value. The Treasury spot curve is:

Year	Spot rate
1	3.20%
2	3.90%
3	4.60%
4	5.20%

Compute the Treasury-curve value before adding any credit spread.

Answer. Discount each cash flow at the matching spot rate. $Price = 35/1.032 + 35/1.039^2 + 35/1.046^3 + 1,035/1.052^4 = \941.96 . This is the value if the risky bond required only Treasury spot rates.

1.14 Question 14 (Calculation)

Using the spot rates in Question 13, compute the annual forward rate from year 2 to year 4.

Answer. $1 + f_{2,4} = [(1.052)^4 / (1.039)^2]^{1/2}$. The two-year forward rate beginning in year 2 is 6.52%. It is the break-even two-year rate for years 3 and 4.

1.15 Question 15 (Calculation)

A 5-year high-yield bond has cumulative default probability of 18% and expected recovery of 45% of par. Approximate the annual expected credit loss spread using a simple straight-line method.

Answer. Expected cumulative loss = default probability x loss given default = 18% x (1 - 45%) = 9.90%. Straight-line annual compensation is about 1.98%, or 198 bps, before liquidity, risk premium, and timing effects.

1.16 Question 16 (Calculation)

An investor in the 37% tax bracket compares a 5.60% taxable corporate bond with a 3.80% tax-exempt municipal bond. Which has the higher after-tax yield?

Answer. Corporate after-tax yield = 5.60% x (1 - 37%) = 3.53%. The municipal yield is 3.80% tax exempt. The municipal bond has the higher after-tax yield by 27 bps, before considering credit and liquidity differences.

1.17 Question 17 (Calculation)

A Treasury note position has \$1,000,000 face value, quoted clean at 100-16+, and semiannual coupon payment of \$22,500. Settlement is 42 days into a 182-day coupon period. Compute full settlement price.

Answer. 100-16+ = 100 + 16/32 + 1/64 = 100.515625% of par. Clean dollars = \$1,005,156.25. Accrued interest = \$22,500 x 42/182 = \$5,192.31. Full price = \$1,010,348.56.

1.18 Question 18 (Calculation)

A new agency mortgage pool follows 100% PSA for the first six months. Beginning balance is \$500,000, and scheduled principal is \$900 per month. Under 100% PSA, CPR is 0.2% in month 1, 0.4% in month 2, and rises by 0.2% monthly. Compute the month-6 ending balance.

Assume prepayment each month is $SMM \times (\text{beginning balance} - \text{scheduled principal})$.

Answer. Applying the PSA ramp monthly gives an ending balance of approximately \$492,859.15 after month 6. The balance declines by scheduled principal plus increasing prepayments as CPR ramps up.

1.19 Question 19 (Calculation)

An investor buys a bond for \$990, receives a \$40 coupon after one year, reinvests nothing because the horizon is one year, and sells the bond for \$1,010. Compute one-year horizon return.

Answer. Total dollar return = coupon + price change = $\$40 + (\$1,010 - \$990) = \60 . Horizon return = $\$60/\$990 = 7.27\%$. This differs from YTM because it uses the realized sale price and actual horizon.

1.20 Question 20 (Calculation)

A leveraged fund buys \$10,000,000 of agency debt yielding 5.20%, finances it in repo at 4.40%, pays 20 bps annual operating/friction cost on assets, and faces an 8% haircut. Approximate return on equity before price changes.

Answer. Net carry dollars = $(5.20\% - 4.40\% - 0.20\%) \times \$10,000,000 = \$60,000$. Equity tied up by the haircut is $8\% \times \$10,000,000 = \$800,000$. Approximate ROE = $\$60,000/\$800,000 = 7.50\%$. The number is attractive only if financing remains available and spreads do not widen.

1.21 Question 21 (Spreadsheet)

Open `hard_curve_trade.csv`. The portfolio manager expects a bear flattening: 2-year rates +80 bps, 5-year rates +55 bps, 10-year rates +25 bps, and 30-year rates +10 bps. Use the key-rate dollar durations to estimate each portfolio's dollar P&L and choose the less exposed portfolio.

Answer. Multiply each key-rate dollar duration by the rate shock in basis points and sum the negatives. Portfolio Alpha loses about \$972,500; Portfolio Beta loses about \$755,000. Beta

is less exposed because it has less short/intermediate key-rate exposure where the shock is largest.

1.22 Question 22 (Spreadsheet)

Open `hard_callable_vs_noncallable.csv`. Using the scenario prices, compute effective duration for the callable and noncallable bonds. Explain the difference.

Answer. Effective duration = $(P_{\text{down}} - P_{\text{up}}) / (2 \times P_0 \times \text{shock})$. The noncallable bond has duration about 7.52, while the callable bond has duration about 3.52. The callable bond's upside price is capped when rates fall, so its effective duration and convexity are lower around the call region.

1.23 Question 23 (Spreadsheet)

Open `hard_credit_expected_loss.csv`. Compute expected loss in dollars for each bond, loss-adjusted spread in bps, and identify which bond has the best spread per unit of expected loss.

Answer. Expected loss = exposure $\times$ default probability $\times$ (1 - recovery). The approximate expected losses are \$132,000, \$112,000, and \$135,000. Loss-adjusted spread is quoted spread minus annualized expected loss as a percent of exposure; Bond C is strongest on this simple measure because its quoted spread remains much wider after subtracting expected loss. This does not mean Bond C is automatically best, because tail risk, liquidity, and downgrade risk still matter.

1.24 Question 24 (Spreadsheet)

Open `hard_mbs_waterfall.csv`. Use the mortgage-pool data to compute SMM, prepayment, total principal, ending balance, and average life contribution for each month.

Answer. The method is month-by-month: convert CPR to SMM, compute prepayment after scheduled principal, total principal, and ending balance. Summing month $\times$ total principal divided by original balance gives the partial average-life contribution for the first six months, approximately 0.014 years; most principal remains outstanding beyond the displayed horizon.

1.25 Question 25 (Spreadsheet)

Open `hard_immunization.csv`. Use the asset and liability cash-flow tables to compute present value and Macaulay duration at the given spot rates. Determine whether the asset portfolio is immunized.

Answer. Discount each cash flow using the matching spot rate. The asset PV is about \$10.15 million and asset Macaulay duration is about 4.55 years. Liability PV is about \$9.60 million and liability duration is about 5.10 years. The portfolio has more present value than the liabilities but is under-duration relative to them, so it is not fully immunized against small parallel shifts.